



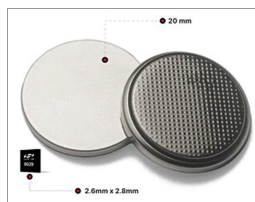
companies and organisations developing or operating computing systems used in AI and machine learning, and for use in supercomputing centres, research institutions, and companies working on AI accelerators or ASICs that require fast data processing.

SK hynix

<https://www.skhynix.com>

Compact Wireless SoC with Memory and Power

Silicon Labs has introduced the BG29 Series 2 wireless SoC family. These chips are built for Bluetooth low energy (LE) devices that require performance



and a small footprint—used in wearables, medical devices, asset trackers, and sensors. The

SoCs include 1MB Flash and 256kB RAM, supporting features like real-time data processing and communication protocols. Reducing size often leads to performance or battery life tradeoffs as medical devices get smaller. BG29 addresses this by combining small size, connectivity, processing capability, and battery efficiency, which are suitable for devices like blood glucose monitors and patches.

Silicon Labs

<https://www.silabs.com>

World's Smallest PPG Sensor Head

SCIVAX Corporation and TSLC Corporation have jointly developed the world's smallest PPG (PhotoPlethysmoGraphy) sensor head. The



sensor head has a mounting area of 1 square millimetre—one-tenth the size of conventional models—with about half the thickness, allowing for

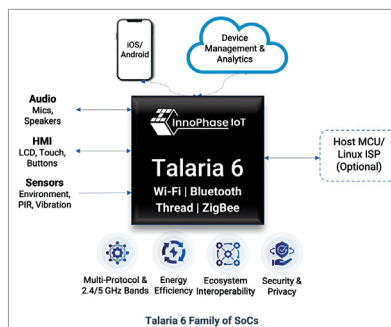
miniaturisation. The PPG sensor head offers several advantages: it allows installation in tight spaces where conventional sensors cannot fit; it enables multiple sensor heads to be arranged in arrays for more precise sensing and the ability to measure blood flow in small areas, supporting use in beauty, home healthcare, and elder care; and its integrated resin package, combining the light source and optical sensor, provides cost efficiency compared to traditional models.

SCIVAX Corporation and TSLC Corporation

<https://www.scivax.com/en> and <https://www.tslc.com.tw>

SoCs Featuring Wi-Fi 6, Multi-Protocol Connectivity

InnoPhase IoT Inc. has launched the Talaria 6 Family of system-on-chip (SoCs). Designed to meet the



demands of the IoT market, these chips integrate multi-protocol wireless connectivity, including Wi-Fi 6, Bluetooth 6.0, Thread, and Zigbee, along with PSA Level 2 and Level 3 security and an ARM Cortex M33 processor. The Talaria 6 family builds on the platform's low power capabilities, providing performance, energy efficiency, security, edge AI processing, and connectivity for IoT devices. Target markets for the SoCs include smart homes, building automation, industrial IoT, and healthcare sectors, with applications across video cameras, doorbells, locks, thermostats, lighting, environmental sensors, appliances, and more.

InnoPhase IoT

<https://innophaseiot.com>

First 'Amplifier-less' MEMS Tweeter

xMEMS Labs has announced Lassen, a micro tweeter designed to bring micro fidelity (μ Fidelity) audio to consumer earbuds. Based on industry feedback



and reviews of its speaker technology, xMEMS developed Lassen as a lower-cost solu-

tion for true wireless stereo (TWS) earbuds. Lassen delivers up to 115dB SPL in the 6kHz to 20kHz range using a 1Vrms audio signal, removing the need for an external amplifier. Lassen delivers audio performance that exceeds older tweeter technologies with dimensions of 3.2 x 5 x 1.15mm and low power use. By removing the amplifier, the company claims to reduce the cost of tweeter integration by about 25%, placing its MEMS speakers alongside other micro tweeter types.

xMEMS Labs

<https://xmems.com>

Smart Power IC for Longer Battery Life

Nordic Semiconductor's nPM2100 power management IC (PMIC) includes battery-powered products' energy-saving and system management features. It fills the gap between simple regulators and complete power management systems,



allowing designers to extend battery life or reduce battery size. Its 35nA ship mode lets devices ship with batteries installed and acts as an on/

off switch. The < 200nA hibernate mode supports wakeups from seconds to days, increasing standby life by up to 3x for devices that wake rarely. A 150nA boost converter reaches up to 90.5% efficiency at 10 μ A and 95% at higher currents. It can be used by developers working on wearables, medical sensors, wireless industrial sensors, remote monitoring devices, and asset trackers.

Nordic Semiconductor

<https://www.nordicsemi.com>